Project Aevum — Phase 16.4 Freeze Record

VERIFIED & FROZEN 🔒

Longitudinal Dynamics: Satiation, Fatigue, Dormancy & Resurgence

MILESTONE STATUS	PHASE 16.4 GOLDEN TESTS	DESKTOP FRONTEND	DATE & AUTHORITY
FROZEN (Gate 0 + Implementation)	16 Golden IDs / 24 Test Cases (100% Green)	0 Errors (Vite + TS Production Build)	2026-10-04 / Freeze Gate Passed

1. Executive Summary & Core Epistemic Quad

Phase 16.4 formalizes the companion's temporal engagement dynamics without reducing human-like longitudinal fatigue and dormancy to hidden scalar scores or destructive lifecycle downgrades. Activity state is an **observational interpretation of temporal participation**, not a loss of personality, forgetting of experiences, or erasure of taste.

Phase 16.4 Epistemic Tenets Enforced:

- `Activity State is Interpretation, Not Identity` (Dormancy does not alter who Aevum is)
- `Dormancy ≠ Forgetting` (Zero degradation of memories, familiarity, or experience count)
- `Satiation ≠ Dislike` (Fatigue from repeated sessions never alters formed preferences)
- `Resurgence ≠ Experience` (Awakening a dormant interest mints 0 experiences and 0 appraisals)

2. The Five Constitutional Locks (GC-413 – GC-417)

Lock	ID	Constitutional Contract	Verification Proof
Session-Unit Fatigue Integrity	GC-413	Session (`calendar_session_id`) is the atomic counting unit. Multiple appraisals within one session collapse into one outcome. An intervening session on another facet resets streak.	G53 (Pattern 1 monotony), G54 (Pattern 2 density), G55 (streak reset on interruption)
Semantic-Time-Only Evaluation	GC-414	Longitudinal lapse is strictly measured via Phase 8C cognitive ticks (integer calendar days). Database `created_at` and wall-clock datetime have absolute zero semantic authority.	G56 (60-day formed threshold), G56b (45-day unformed threshold), G65c (zero float fields)
Activity-State Non-Destructiveness	GC-415	Entering `SATIATED` or `DORMANT` is read-only projection: zero mutations, deletes, or downgrades to `character_experiences`, `character_facet_appraisals`, or `character_facet_preferences`.	G57 (Dormant snapshot before == after), G58 (Satiated snapshot before == after)
Resurgence ≠ Experience	GC-416	`ResurgenceEvent` only logs the awakenment of a dormant domain to `character_resurgence_events`. It produces 0 experience records, 0 appraisals, and 0 preference deltas.	G62 (delta = 0 on experience, appraisal, and preference tables; exactly 1 resurgence row)
User-Reported Zero Refresh Authority	GC-417	Only first-person `DIRECT_INTERACTION` and co-present `SHARED_INTERACTION` refresh Aevum's active anchor. `USER_REPORTED` and `OBSERVED_MEDIA` carry zero refresh authority.	G64 (categorical allowlist inspection), G64b (USER_REPORTED omitted from eligible query)

3. Satiation Patterns & Recovery Asymmetry

Discrete Pattern Formalization (Zero Floats – Permanent Note A)

- **Pattern 1 (Acute Monotony)**: ≥ 4 consecutive sessions on the exact same facet (ordered by `timeline_tick ASC`) AND ≥ 2 fatigue appraisals (`TEDIOUS_EXHAUSTING`) in the trailing 3 sessions.
- **Pattern 2 (Hyper-Density)**: ≥ 7 distinct sessions within 7 Phase 8C calendar days AND ≥ 3 fatigue appraisals in trailing 5 sessions.
- **Diversity Anti-Fatigue Invariant (G52)**: ≥ 4 distinct sub-facets engaged within the 7-day window unconditionally blocks Pattern 1 (Acute Monotony) via set cardinality.

Recovery Asymmetry: Satiation Cooldown vs. Resurgence Awakening

A critical architectural boundary enforced in Phase 16.4 is the strict separation between **satiation fatigue relief** and **dormancy resurgence**:

```
SATIATED ──[ Natural Cooldown: ≥ 3 Phase 8C days with 0 domain sessions ]──────────────> ACTIVE
SATIATED ──[ Time Lapse: ≥ 60 Phase 8C days without engagement (Fatigue Healed) ]──────> DORMANT
DORMANT  ──[ Valid Anchored ResurgenceEvent (conversational cue, media cue, etc.) ]────> ACTIVE
```

> **Strict Semantic Rule:** External stimuli *cannot* forcibly bypass the satiation fatigue cooldown. Calling `validate_resurgence_event()` on a `SATIATED` domain raises `InvalidResurgenceStateError` (G65b). Resurgence is exclusively a Dormancy recovery mechanism.

4. Storage Deletion Semantics & Immutability Audit

System Character Deletion Policy

• **Production In-Lifecycle Deletion**: Physical deletion of character definitions is strictly prohibited and outside the production runtime. Zero API routes, controller methods, or services implement physical deletion of `character_definitions`.
• **SQLite Foreign Key CASCADE**: `ForeignKey("character_definitions.id", ondelete="CASCADE")` is consistently applied across all Aevum Core tables to support clean tenant / testing database teardown without orphaned rows.
• **ORM-Level Append-Only Log Protection**: `CharacterResurgenceEventTable` is fortified by SQLAlchemy ORM event listeners (`before_update` and `before_delete`), raising `ResurgenceMutationProhibitedError` on any modification or deletion attempt (verified in G63).

5. Architecture Seam & Isolation Contracts

• **Pure Separation of Concerns**: `dynamics_models.py` contains pure DTOs, enums, frozen integer thresholds, and pure evaluation functions testable without database access.
• **Read-Only Service Pipeline**: `LongitudinalDynamicsService` reads Phase 16.1 experiences and 16.3 appraisals/preferences to compile discrete session contexts; it performs zero writes during activity evaluation.
• **Single New DB Table**: Phase 16.4 introduces exactly one append-only table: `character_resurgence_events`. Zero schema modifications to prior frozen tables.
• **Phase 17 Capability Lockout**: Autonomous agency, background interest exploration, and proactive scheduling remain strictly prohibited (raising `Phase17ProhibitedError` if invoked).

6. Complete Verification Audit Matrix

Verification Suite	Scope & IDs	Tests Executed	Result
Phase 16.4 Dedicated Golden Suite	G50 – G65 (16 Golden IDs)	24 / 24 test cases	**PASS** (100% Green, 0.47s)
Phase 16 Master Suite	Phase 16.1 + 16.2 + 16.3 + 16.4	103 / 103 test cases	**PASS** (100% Green, 1.77s)
Full Monolithic Backend Regression	Full Aevum Core Test Suite	2,553 / 2,553 test cases	**PASS** (100% Green, 0 Regressions)
Desktop Frontend Production Build	TypeScript + Vite build	45 modules compiled	**PASS** (0 Errors, 902ms)
Static AST Boundary & Float Audit	Phase 16.4 codebase scan	Zero float literals, 0 forbidden imports	**PASS** (0 Violations)

7. Roadmap Status & Next Milestone

Phase	Milestone Name	Status	Core Focus & Contract
16.1	Experiential Record & Provenance	FROZEN 🔒	Append-only experiential events, 7-tier provenance hierarchy
16.2	Interest Domain State Machine	FROZEN 🔒	Deterministic lifecycle derivation, curiosity state machine
16.3	Facet Appraisal & Preference Formation	FROZEN 🔒	Episodic facet appraisal, taste projection, permutation invariance
16.4	**Longitudinal Dynamics: Satiation & Dormancy**	**FROZEN** 🔒	**Acute monotony, hyper-density, non-destructive dormancy, resurgence**
16.5	Personal Growth Context Injection	NEXT →	Deterministic XML injection of personal growth context into prompt pipeline
17.0	Autonomous Agency & Curiosity	PLANNED	Autonomous companion agency during background scheduling loops